

Gauraj Rakholiya

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Education

L.D.R.P. Institute of Technology

B.Tech. in Computer Science
CPI : 8.34/10

September 2021 – April 2025

Gandhinagar, Gujarat

P.P. Savani Vidyabhavan

Gujarat Secondary and Higher Secondary Education Board
Percentage : 86%

June 2019 – April 2021

Surat, Gujarat

P.P. Savani Vidyabhavan

Gujarat Secondary Education Board
Percentage : 86.83%

June 2018 – April 2019

Surat, Gujarat

Work Experience

Backend Developer

-Vrutti Technologies Pvt. Ltd. (February 2025 – Present)

Animeo IP by Somfy | [View Project](#)

- Developed a scalable NestJS-based backend for a home automation platform integrating IoT device control using MQTT and real-time updates via WebSockets/Socket.io.
- Built the real-time device control plane: a Socket.IO gateway with 104 handlers across ~200 events and 20 rooms, streaming live motor positions, sensor telemetry, GPIO events, and discovery progress between embedded controllers and browser clients.
- Designed a data-driven RBAC system (38 modules, 128 permission rows, 5 role tiers) enforced across ~200 endpoints with row-level ownership checks.
- Built a versioned migration engine for embedded SQLite (12 schema + 10 data migrations, idempotent seeders) with transactional project import including automatic backup and restore-on-failure
- Implemented pull-based OTA firmware distribution with semver compatibility handshake, automatic update/downgrade, live progress streaming, and rollback via backup-restore.
- Containerized services with Docker Compose for ARM64 embedded Linux targets, achieving dev-production parity and reproducible deployments on field controllers.

Content Management System | [View Project](#)

- Built and maintained a production Node.js/TypeScript/Express CMS backend exposing ~708 REST endpoints across 131 modules and 95 Sequelize/MySQL models, powering both a public mobile/web client and an internal admin console.
- Implemented JWT authentication with a role-module-permission RBAC matrix supporting per-verb row-level scoping, full audit logging of mutating requests with before/after snapshots.
- Designed a media pipeline handling upload → optimization → CDN delivery: Multer validation, Sharp derivatives, TinyPNG compression, BlurHash placeholders, and dual-bucket AWS S3 storage with presigned URLs — tuned libvips concurrency.
- Centralized publish/unpublish/reject workflows into one config-driven status engine covering 30+ content models, replacing per-module handlers; paired it with an automated publish-expiry sweep that retires stale content daily without editor intervention.
- Built a database-driven cron scheduler whose jobs (YouTube channel sync, daily darshan release, sitemap regeneration, log rotation) are added, retimed, or paused at runtime with no redeploy, with 3-attempt retry and per-run execution logging.

Technical skills

Programming Languages: C++, C, Python, Java, JavaScript, SQL, MySQL, SQLite

Libraries and Frameworks: React.js, Node.js, Bootstrap, NestJS

Development Tools and Methodologies: Git, GitHub, Docker, API Integration, AWS S3, MQTT

Interpersonal Skills: Team Collaboration, Project Management

Academic and Extracurricular Achievements

- **Finalist** - Student Startup and Innovation Policy Hackathon 2023.
- Completed a course in Programming in Java through NPTEL-Swayam.